

Pre-Thesis I Report

Classifying and Predicting Zero-Day Attacks with AI

by

Saumik Das Turja
22101090

Md. Meheraj Hossain
21301751

Maisha Fairouz
21201031

Sibgatullah Tasnim
22101428

Ayesha Ammarah Ashrafi
22101258

A thesis submitted to the Department of Computer Science and Engineering
in partial fulfillment of the requirements for the degree of
B.Sc. in Computer Science and Engineering

Department of Computer Science and Engineering
Brac University
February 2025

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Declaration

It is hereby declared that

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3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

Student's Full Name & Signature:

Saumik Das Turja

Saumik Das Turja

22101090

Meheraj

Md. Meheraj Hossain

21301751

Maisha Fairouz

Maisha Fairouz

21201031

Sibgatullah

Sibgatullah Tasnim

22101428



Ayesha Ammarah Ashrafi

22101258

Approval

The thesis/project titled “Classifying and predicting Zero days attack with AI” submitted by

1. Saumik Das Turja (22101090)
2. Md. Meheraj Hossain (21301751)
3. Maisha Fairouz (21201031)
4. Sibgatullah Tasnim (22101428)
5. Ayesha Ammarah Ashrafi (22101258)

of Fall, 2024 has been accepted as satisfactory in partial fulfillment of the requirement for the degree of B.Sc. in Computer Science in February 1, 2025.

Examining Committee:

Supervisor:
(Member)



Muhammad Iqbal Hossain, PhD

Associate Professor
Department of Computer Science and Engineering
Brac University

Abstract

For business firms and establishments alike Zero day threats have become a growing concern. They are particularly dangerous and difficult to detect since they generally have no recognizable attack signatures; therefore, current security structures that are mostly signature based ML models fail to counter these threats. Furthermore, they target previously unknown vulnerabilities in the system. The system remains susceptible to damage until a patch is released. Because the attacks occur post-deployment and vendors have no prior knowledge of the problem, preparing a patch or an immediate security solution can become difficult. Times like this, it becomes vital to defend the system against attack and prevent further compromise. This thesis tries to design a multilayer hybrid attack detection system for detecting ZDAs aiming to decrease false positive rates and increase accuracy. Our system will use honeypots, develop adaptability to new exploits and have an added XAI layer for faster and more insightful response.

Keywords: Zero-Day Attacks, Machine Learning, Deep Q-Networks, Signature-Based Security Structures, False Positive Rates, Honeypots , Explainable AI (XAI), Reinforcement learning.

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Chapter 1

Introduction

1.1 Background

With the globalised world of today, cyber attacks are among the most dangerous and ongoing threats, endangering people, businesses and even governments. Zero-day attacks exploit issues in software, hardware that are unknown to programmers and security experts. It is the most dangerous type of attack. These vulnerabilities leave developers with “zero days” to create and deploy a patch. Companies often become victims of financial losses, sensitive data violations, and operational disruptions. Traditional detection systems based on signatures struggle to keep pace with the fast-evolving patterns. Attackers frequently modify these patterns, making them challenging to identify and stop[7] [20]. To solve this, we need advanced and malleable detection techniques.

Honeypots are planned to attract and monitor attackers. Honeypots trick attackers into interacting with fake environments, capturing their tactics and techniques[8]. Honeypots become even more effective when combined with technologies like machine and reinforcement learning. These hybrid approaches can recognize abnormal behavior in network traffic and adjust their detection strategies dynamically in real time[7]. Anomaly-based systems, which focus on identifying behavior and offer an alternative but are often plagued by high false-positive rates. These false positives flood cybersecurity analysts with excessive alerts, making it harder for them to work efficiently and undermining the reliability of these systems in real-world use[20]. Nowadays, attackers employ sophisticated methods like code mutation, encryption, and anti-analysis mechanisms to evade detection. Identifying zero-day attacks becomes much more difficult with these advanced techniques[1] [8].

Reinforcement learning (RL) is able to scan and process enormous volumes of data by seeking out concealed trends and patterns. Proximal Policy Optimization and Deep Q-Networks empower the systems in responding to unknown attack patterns; this is considered a limitation shared by all types of static pattern detection systems[22]. Using Explainable Artificial Intelligence (XAI) makes detection systems easier to understand. This helps security teams trust automated tools and respond quickly to attacks[12]. This research focuses on combining honeypots, reinforcement learning, and XAI to create a powerful framework for detecting zero-day attacks.

1.2 Rational of the Study or Motivation

Zero day attacks highlight gaps in existing cybersecurity frameworks. Traditional approaches, such as signature-based IDSs are limited, as they depend on present data and cannot cope with new, unpredictable attacks. These difficulties highlight the overwhelming demand for flexible and versatile solutions that can adapt to the constantly changing face of contemporary cyber-attacks. Attackers use advanced techniques to bypass static defenses, highlighting the need for efficient solution [3].

To overcome these challenges we can use hybrid systems. A Hybrid system combines machine learning, reinforcement learning, and honeypots. Reinforcement learning offers the ability to adapt and respond dynamically to new attacks, while Explainable AI ensures that these systems are more transparent and actionable for cybersecurity professionals[20] [8]. This research helps by suggesting a method of detecting zeroday attacks more efficiently and accurately through the combination of AI and data derived from honeypots.

This research is important because it has the ability to fill the distinction between latest advancements in AI and its real-world use in cybersecurity. The proposed framework that combines honeypots, reinforcement learning and XAI, solves the main shortcomings of current systems by lowering false positives and negatives. The most important objectives of this research is to optimize the accuracy of detection and develop a stronger cybersecurity system.

1.3 Problem Statement

Zero days attacks are one of the biggest problems in modern world. Attackers can infiltrate and hijack a network before there is a solution or update. These are commonly known as zero-day attacks, which characterize where such forms of attacks exploit any unannounced vulnerability within the systems and software. Traditional security measures dependent on known threat signatures cannot recognize such attacks, making them even riskier. Software systems are becoming complex, and more devices are connected, so zero-day attacks occur more frequently with higher consequences. It causes breaches of information, financial losses, and problems in processes[13].

The most challenging task, however, is to find and stop these kinds of attacks in real time, as many of them just appear like everyday activities, hence difficult to identify. Fake tracking systems and systems built on signatures that are already in use are unable to keep up with how zero-day threats change and increase. Zero-day attacks are extremely hazardous because they exploit undisclosed flaws and put organizations and humans in great danger. Such an attack has the possibility to disclose huge quantities of data, economic damages, and destroy key infrastructures. These types of threats cannot be detected by traditional security systems; therefore, zero-day techniques remain a very serious problem in cybersecurity[24] [1]. The solutions to these attacks build confidence in digital systems by protecting sensitive

information and reducing how often tasks are disrupted. Locating and mitigating zero-day attacks becomes a major priority toward good cybersecurity, ensuring the security of data and minimizing the threats of dynamic and ever-evolving cybersecurity challenges.

1.4 Objective

This research paper is based on the development of an efficient AI-based framework that identifies zero-day attacks with much accuracy and with minimal time consumption.

- Assess the effectiveness of honeypot deployment for logging and tracking certain attacker activities[7].
- Observe how well the machine learning models classify safe traffic from potentially dangerous patterns.
- Test the system to ensure it learns from past cyber-attacks and handles new threats efficiently.
- Create reports with actionable, clear steps identified to help build trust and efficiency.
- Train the model using the datasets to figure out how effectively it can recognize and prevent known cyberthreats[25].

These will in turn lead to an integrated robust AI-driven solution which will help in improving the security measures against zero-day attacks.

1.5 Methodology in Brief

The proposed approach takes advantage of the systematic and multi-step procedure in identifying, examining, and responding to novel attacks on cybersecurity, specifically zeroday attacks on IoT networks. The procedure combines supervised machine learning, honeypot-based threat detection, Siamese network based anomaly comparison, and future incorporation of adaptive defense based on reinforcement learning and explainable AI.

The pipeline starts with attack model training on credible attack data sets and subsequent real-time anomaly detection against suspicious behavior patterns, which

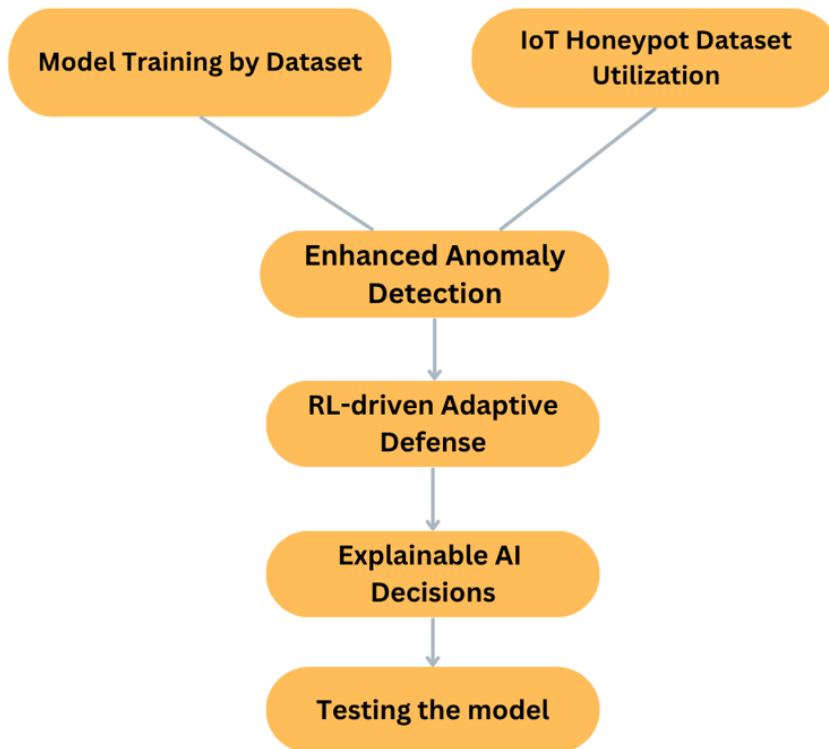


Figure 1.1: Work Flow

is fine-tuned by a comparison with activity that is being monitored by honeypots. This leaves us with a list of infrequent anomalies that are good zero-day candidate threats. The whole pipeline is modular and adaptive and includes an ultimate stringency test phase where the system is tested against known and simulated attacks.

Workflow Explained

Model Training by dataset:

We initially train supervised classifiers (Random Forest, XGBoost, CatBoost, MLP) on a labeled dataset (e.g., CICIDS2023). These models learn to recognize benign and malicious traffic based on previous patterns and structured features. The models the already trained ones are then used for anomaly detection in new data.

IoT Honeypot based Data Collection:

A honeypot system is used exclusively to lure in bad traffic and record real-world attacker behavior. All activity such as commands issued, system calls, and network traffic is recorded by the honeypot. All information is normalized and ready for comparative analysis.

Enhanced Anomalies Detection: The supervised model-identified anomalies are further compared to honeypot data using a distance-matching mechanism through Siamese networks. Manhattan distances are calculated and the anomalies which are different from honeypot patterns are removed—such strange anomalies are great indicators of probable zero-day attacks.

RL- Driven Adaptive Defense: In the future, we apply reinforcement learning methods such as Deep Q-Networks (DQN) and Proximal Policy Optimization (PPO). These will allow the system to learn defense strategies on its own from known threats detected ahead of time, blocking evil IPs and reacting to emerging attack trends.

Explainable AI Decisions: To add to confidence and openness, we will incorporate Explainable AI (XAI) mechanisms. They will provide explanations in terms of the users such that they can see why certain anomalies were detected, what their likely effects were, and the steps to mitigate them.

Testing the Model: The entire system is evaluated with standard attack datasets (e.g., CICAPT-IIoT) and synthetically produced zero-day attacks through the use of Generative Adversarial Networks (GANs). This will render the system immune to known as well as unknown threats.

1.6 Scopes and Challenges

Identifying and preventing zero-day attacks remains amongst the most difficult problems in the field of cybersecurity. There has never been a better chance to improve security protocols and lower the dangers of these advanced attacks. To implement these kinds of systems, we have to consider various kinds of technical, operational and ethical aspects. This section contains scope where our proposed solution can contribute impactfully and challenges that we have to solve to deploy our system successfully and improve its effectiveness.

Scopes :

- **Enhanced detection capabilities** : Two dataset comparison using siamese network, the suggested approach can significantly enhance the detection and prevention of zero day attacks.
- **Improved accuracy** : Our proposed system will reduce false positive and negative rate by integrating explainable AI and hybrid anomaly detection.
- **Real time response** : Reinforcement learning will enable the system to adapt and response to threats in real time.

Challenges :

- **High computational cost** : These kinds of hybrid AI driven systems need high computational resources and costs.
- **Data quality and noise** : Getting high quality data for training our AI model is a big challenge. Incorrect or noise data can decrease system performance
- **Scalability Issues**:The scalability problem arises when artificial intelligence-based systems are integrated into large networks because maintaining huge volumes of data, organizing the traffic pattern, and ensuring quick replies is a computationally intensive task that requires advanced algorithms.
- **Ethical and Privacy Issues**: Collection and analysis of user data does raise ethical and privacy issues about invading an individual's freedom in training and monitoring.
- **Evolving threat landscape**:Since the system has to be constantly successful in reactions to new techniques of the attack, approaches and models need to be revised all the time.

Chapter 2

Literature Review

2.1 Preliminaries

Honeypot: Honeypot is employed to identify unauthorized use, gathering information on attack profiles and malware. It analyzes attackers behaviors and strengths system defenses.

Reinforcement Learning (RL) : A part of Machine Learning (ML) that can speak with its environment and train software in an attempt to make a decision in a quest for the best outcome. Deep learning is combined with Q-learning to develop techniques such as Deep Q-Networks (DQN) which are particularly effective in handling zero-day attacks by learning a policy to maximize rewards over time. One of the reinforcement learning algorithms is Proximal Policy Optimization (PPO). PPO is employed for its stability and efficiency in training.

Anomaly Detection: It is a tool for recognizing actions or patterns that have extreme deviation in behavior patterns. It is used for finding security vulnerabilities or suspicious behavior. k-Nearest Neighbors (KNN) is a Supervised Learning Algorithm most commonly used in the work of classification and regression. Isolation Forest is an extension of a decision-tree technique-based Unsupervised Learning Algorithm with an added feature to generate a variety of decision trees in an attempt to detect outliers in information.

Intrusion Detection and Prevention Systems (IDPS): IDPS are security software capable of discovering malignant behavior in a system through observation of network activity. When a threat is suspected, they inform system administrators with immediate blocking or mitigation of the coming attack in order to ensure integrity within the network.

Generative Adversarial Networks (GANs): Two neural networks in a generative adversarial network compete with one another using deep algorithms in a quest to make ever-improved guesses. Generative Adversarial Networks, or GANs, are AI algorithms consisting of an algorithm generating artificial information and an evaluator rating information. Because they're competing and improving one another,

this generates real output, such as work, photos, or attack simulations.

Random Forest : There is a supervised learning techniques named Random Forest is used in both regression and classification tasks. For the performance improvement and for the reduction of overfitting, it will build multiple decision trees on different random sub-sample of data set and random subset of the attributes, aggregate the predictions of these trees to reach the final prediction (majority or weighted average). In several implementations, this joint treatment leads to reliable, efficient results over a range of applications.

XGBoost: A classification and regression supervised learning technique is referred to as XGBoost. It constructs a series of decision trees through gradient-boosting techniques, each one correcting the mistakes of those constructed before it. XGBoost is highly reputed for its speed and correctness, and it performs exceptionally well on structured tabular data.

CatBoost : CatBoost is a machine learning algorithm for supervised learning that performs effectively on both classification and regression tasks. It reduces human encoding by making optimization effective in processing categorical features. CatBoost, as a gradient boosting algorithm, is capable of dealing with structured and real-world data because it is effective in generalization, has efficient training, and provides high accuracy.

MLP Classifier : Developed from artificial neural networks, the MLP (Multilayer Perceptron) Classifier is a supervised learning algorithm. The above architecture contains an input level, one or more levels that are invisible, and an outcome level. Every completely integrated level uses non-linear activation mechanisms to learn complex patterns. MLP is trained using back-propagation, which suits classification problems with text, images, or structured data. It functions optimally when the inter-relationships among features are non-linear and complicated.

Siamese Network : One of the neural network architectures that is used extensively in supervised learning, particularly for tasks such as verification of signatures and face recognition, is called a Siamese Network. It is used to compare two samples of inputs at the same time and comprises two smaller networks with the same weights. The network is best suited for comparing pairwise tasks or one-shot learning with sparse information since it learns to score similarities by comparing feature embedded data.

Zero-Day Attacks:Zero-day attacks will continue using completely unknown bugs in the program until such a time that the programmers update it or repair the bug. This, therefore, makes the factor of risk high with no earlier warnings and security accorded. This also demands rather tough ways of detection as well as mitigation processes.

2.2 Review of Existing Research

Brinkley et al. [16](2024), in their research, utilized machine learning methods like Random Forest, SVM, and Neural Networks that were able to detect new, unseen before malicious activity. In their model, they used a dataset with a total of 2850 samples which were collected from public cybersecurity repositories. The dataset was preprocessed in several difficult steps such as data cleaning, normalization, and segmentation for training. Feature extraction techniques like operations, and anomalous network traffic were applied to capture more complex behavioral data. In their study, the Neural Network model performed optimally at 92.4%, followed closely by Random Forest at 91.2%.

Satam et al. [13](2023) in their article give a reflective review of a number methods, such as anomaly detection, machine learning (ML), and hybrid models, to counter zero-day attacks. The integration of tools like VirusTotal and MetaDefender for threat detection is particularly noteworthy, as these tools can inspire the incorporation of external verification mechanisms to enhance RL models' outputs. Furthermore, the paper includes attack simulations such as SQL Injection and Remote Access Trojan, which provide a practical understanding of attacker behavior. These simulations align closely with the network traffic monitoring.

Pitre et al. [7](2022) in their paper they propose a hybrid two-stage IDS. This approach combines feature selection and fine-tuning techniques for improving anomaly detection. This two-stage model uses logistic regression in the first stage and a fine-tuned dataset in the second stage in order to distinguish between false and true positives. The study explores various optimizers such as Adam and RMSProp to increase classification speed and accuracy.

Zaki et al. [24](2024) in their article present IDS (Intrusion detection system) for identifying ZDAs in IoT networks using ensemble machine learning techniques. For feature selection, a mixture of a sparse random projection and a composite model comprising classifiers k-Nearest Neighbors (KNN), XGBoost and Stochastic Gradient Descent (SGD) is utilized by the authors. This method successfully reduces dimensionality while maintaining the most relevant information for model training. Based on the CIC-DDoS2019 and CICIDS2017 datasets, the model records with an impressive 99.91% accuracy and low intrusion detection time of 0.22 seconds, making it feasible for real-time applications. The paper's focus on applying GridSearchCV to optimize each classifier's hyperparameters guarantees improved performance over earlier methods.

Kareem et al. [22](2024) in their paper used Neural Transformers for Zero-Day Detection. The transformer model processes sequences of network packets (size of the packet, protocol type, source, and destination addresses) to learn patterns that are typical of normal traffic and those that are unusual. They used public datasets like CICIDS 2017 to provide diverse examples of attack patterns. They claimed that they attained an accuracy of 96% and precision of 94% in the transformer model, which is much higher than logistic regression, SVM, and random forest. Also, the average latency per packet was 25 milliseconds, which is significantly lower in

comparison to other deep learning models tested.

Abbas-Escribano and Debar [8](2023) proposed an improved high interaction honeypot with an emphasis on having an increased level of interaction with attackers for purposes of detection and analysis of attacks while maintaining system integrity. This improved model consists of virtualized architecture elements positioned at different network levels such as entry, intermediary, and internal. The model derived its basis from the MITRE ATT&CK taxonomy for mapping adversary tactics and techniques. These include brute-force SSH and RDP service attacks, as well as broad scanning activity. These were identified by the recording of this improved honeypot technique for the time span of over 17 days, during which 1.5 million events were recorded. The most prominent finding of this technique is that the attackers were not allowed to successfully penetrate the honeypot's deepest layers, thus the model logged the behavior of these attackers very well, providing valuable Indicators of Compromise (IoCs).

Diloglu [3](2022) suggested the detection of zero-day attacks through deep learning models in the systems of connection, and his model outperformed the traditional detection algorithms for all measures of evaluation such as F-score, accuracy, and detection time. The proposed methods were trained and tested as per the following standard datasets: KDD '99, NSL-KDD, UNSW-NB15, and CIC-IDS2017. The approach was grounded on supervised and unsupervised deep learning algorithms. It included artificial neural networks and autoencoders with batch normalization with the objective of improving the attack detection capabilities. High accuracy was achieved by supervised models, and from the findings, the CIC-IDS-2017 dataset had a 97% accuracy rate, while unsupervised models were poor in classification in older datasets.

Balamurugan, M. [15](2024) attempted to show the extent to which honeypot utilization and artificial intelligence reduces detection and mitigation in zero-day attacks in their paper. This work also emphasizes the privileges of AI and honeypot systems in detecting unknown attacks and preventing lateral movement of threats in computer networks. For this work, supervised and unsupervised machine learning models are being used. The name of the datasets used for this research paper is UNSW-NB15, CICIDS2017. In the case of AI-enhanced honeypots, the detection rate is 92%, whereas there is a 75% detection rate in the traditional systems. Average response time came down from 5% to 2% compared to the conventional systems. The false positive rates have been decreased from 10% to 3%, whereas the false negative rate went down from 8% to 2%.

Igugu, A. [18](2024) in his article talks about the use of machine learning and artificial intelligence methods in cloud infrastructures to identify zero-day-attack detection. In this case, Machine learning algorithms such as Catboost, XGboost, application of neural networks, Support Vector Machine and collective learning methods are utilized. In this article, the AWS-CIC 2018 dataset has been used. Ensemble learning models regularly excelled other algorithms with a success rate of up to 99% for known attacks.

In their academic research work, Nkongolo et al. [6](2022) took the ambitious and

challenging work of developing a new cloud-based methodology for zero-day attack detection, notoriously challenging to detect, and developed it in a novel manner. For developing such an innovative model, they utilized the rich UGRansome1819 dataset, a strong basis for such work. In this particular piece of work, in a novel Genetic Algorithm, with just a single objective in mind, i.e., maximize and enhance performance measure for a range of Machine Learning Algorithms, they used a particular algorithm, i.e., commonly utilized and popular algorithms like the Support Vector Machine, Naive Bayes, and Random Forest, and these algorithms have high regard and usage in the machine learning field, an indication of its usefulness and usage in overcoming challenging computational issues. The cloud services used for building and validation include the S3 bucket and SageMaker by Amazon Web Services. The research shows that accuracy of 99.6% is achieved by the Random Forest model and it outperforms other models on CAIDA and UNSWNB-15 datasets. The research showed some key results. It found that using ensemble learning boosts how well weak classifiers like Support Vector Machine and Naive Bayes perform when paired with the Genetic Algorithm. Genetic Algorithm was also useful in feature selection and in reducing computational time and overfitting by eliminating most of the less important attributes.

Sarhan et al. [12](2023) in their research tried to improve ZDA detection using a novel zero-shot learning (ZSL) framework. The problem with current ML based models is the use of signature-based NIDs. There is a publicly shared documentation called Common Vulnerabilities and Exposures (CVE) list. When a ZDA has been identified it is added to this list. However, ZDAs are inherently novel and if IOCs related to the threat are yet to be discovered, systems secured using signature-based NIDS prove to be highly vulnerable. The paper, therefore, tries to develop a modern, more enhanced security system contrary to that of traditional NIDs to overcome aforementioned limitations in detecting ZDAs. The proposed methodology encompasses three main concepts: 1) Phishing, MitM, Distributed Denial of Service (DDoS), malware attack, cross-site scripting (XSS), and SQL injection are some of the most prevalent attacks for which training samples in terms of sample data can readily be acquired. 2) Zero day attacks are novel attacks which become apparent post-deployment and have no labelled data samples for training. 3) Semantic attributes: The ML model we train in the process understands and learns the core concepts behind some known attacks in such a thorough way such that it can identify new attacks outside its known set of attacks by comparing the identified anomaly to the generalized frame of known attacks. The attributes the model generalized are called semantic attributes. Two machine learning models (MLP and RF) were used, and the performance of each depends on how complex the attack happens to be. Both models were able to detect approximately 95% generic attacks i.e. Backdoor, DoS, Shellcode etc. successfully. However, Reconnaissance, Exploits and Analysis appear more challenging to detect. In this case, each model achieved around 90% Z-DR.

Sayduzzaman et. al [23](2024) in this paper combined interoperability with XAI for detecting ZDA faster. The paper attributes the failure in detecting ZDA to the poly-morphic characteristics of malware, therefore proposes the integration of XAI

for faster, more accurate detection. XAI brings clarity, interpretability and explainability of the results. The paper defines interoperability as the ability of various application platforms including software, hardware to exchange data and share real-time threat intelligence efficiently. The model creates a training dataset which is later used to generate two SHAP (SHapley Additive exPlanations) values via XAI; one to detect attack pattern analysis another for anomaly analysis. In order to create this dataset, the model uses real time threat information. The anomaly can then be detected by sending this output pattern to the IDPS system. The algorithms used here were AdaBoostM1, RandomSubspace, LogitBoost, DecisionTable, Support Vector Machine (SVM), Autoencoders, Decision Tree, Cost Sensitive Pareto Ensemble (CSPE-R), Feature Selection, Support Vector Machine (FSVM), Hitter and Graph Technique. According to the results, AdaBoostM1 and RandomSubspace produced the highest accuracy in case of ZDA detection and anomaly detection respectively. For anomaly detection LogitBoost and DecisionTable, after the use of XAI show least computational time. The paper poses their only limitation: a small dataset. Other than that, the proposed methodology, according to results, has been shown to detect any attack, including zero-day attack.

Hairab et al [4](2022) clarifies utilizing CNNs classifier to identify unknown malicious attack traffic. Various regularization techniques like L1 and L2 have been used to avoid overfitting and control the complexity of the classifier. Various attacks data like Service Scanning, Data Theft, Key Logging, OS Fingerprinting, DoS and DDoS are present in this dataset. They work better on all the metric evaluations when applying regularization methods as opposed to traditional CNN. They trained, tested, and cross-checked classifiers in the aforementioned cases on the Bot-IoT dataset. This dataset holds various attacks data such as Service Scanning, Data Theft, Key Logging, OS Fingerprinting, DoS and DDoS. They achieve their best 99.98% accuracy with CNN L2.

Mbona et al [5] (2022) demonstrates that semi-supervised ML methods are efficient in zero-day attack detection when important features are selected optimally. They report the best performance by applying one-class support vector machines to zero-day attack detection with an F1 score of 85%. Four datasets were utilized for this article to examine zero-day network intrusion attacks such as UNSW-NB15, CICDDOS2019, OTINTRUSION2020, CIRA-CIC-DOHBRW-2020. They opined that the law of Anomalous numbers, or Benford’s law, can be used to determine such features that tend to differ drastically in zero-day and ordinary network traffic.

Kandhro et al. [11](2023) in their paper they’ve written about the use of deep learning models, such as CNNs, RNNs, and autoencoders, to detect malicious intrusions in IoT-driven environments. Their study concentrated on how well traditional intrusion detection systems detect online threats, particularly zero-day attacks. They trained algorithms on datasets such as NSL-KDD and UNSW-NB15. They were able to classify both common and uncommon network threats and get high accuracy. It uses real-time classification pipelines. However, the absence of a specific industrial environment and testing against honeypot logs limit this research’s potential. We

fill the gap by combining anomaly detection with real flows produced by honeypots.

Faldi et al. [9] (2023) implemented a honeypot-based server security system to analyse how different types of cyberattacks interact with network defense systems. Honeypots are particularly effective at detecting application-layer attacks, and can provide valuable early-warning data when we are using signature-based IDS with it. They placed honeypots to monitor attacks such as GoldenEye, LOIC and Slowloris by using pfSense and Snort IDS in a controlled lab environment. A key strength of their setup was its simplicity and real-time alert capability but there is a major problem. That is, their system can't predict new attack patterns, especially those not previously found. Here, we are referring to not just logging attacker behavior through honeypots, but also using machine learning models to identify new zero-day attacks by anomaly modeling and pattern similarity.

Ali et al [2] (2022) highlighted methods of artificial intelligence for the identification of zero days attack, which are difficult in comparison to conventional detection techniques based on known vulnerabilities. It discusses several attack scenarios with datasets including ICS Gas Pipeline, ISOT, ISCX, NSL-KDD, CIDD, CICIDS2017, SPMD, Enron Spam, UNSW-NB15, Kaggle Microsoft Malware, and SWaT. Decision Trees, Random Forests, MLP-IDS, Bloom Filter, LSTM, HML-IDS, Kalman Filter, Zero-Shot Learning, SVDD, Autoencoder, CNN, Reinforcement Learning, DNN, tDCGAN, WAVED, Grey Wolf Optimiser, Stacker, and MSALSTM-CNN are some of the algorithms considered in this work. Stacker (99.97% on CICIDS2017) and tDCGAN (95.74% on malware datasets) are top performers when evaluated in terms of accuracy, precision, recall, and F1-score.

Topçu et al [14] (2023) provide a novel way of finding zero-day cyber attacks by mining Twitter posts. The approach is based on actual time, public tweets to alert people about problems before these become widespread. The authors have extracted the tweets with the terms zero-day, zeroday, and 0-day. That text is then cleaned, sorted, and mined with a mix of NLTK and TensorFlow tools. NLTK tackles the routine tokenization and basic security checks while TensorFlow quickly turns each tweet into tidy, small vectors that lift overall accuracy. Its efficacy compared to conventional methods is testified by its 80% detection rate with TensorFlow and 60% detection rate with NLTK.

Guo [10] (2022) discusses critically methods that utilize machine learning to identify zero-day attacks, which take advantage of newly found software vulnerabilities and mostly evade conventional security measures. Several datasets have been reviewed in the paper, such as CIC-IDS2017, NSL-KDD, CSE-CIC-IDS2018, and private datasets. The study notes that feature values change widely and that realistic zero-day samples are almost absent. The author reviews a mix of methods, naming supervised and hybrid tools like Random Forest and Support Vector Machines, plus unsupervised options such as One-Class SVM and autoencoders, and finally transfer-learning approaches like manifold alignment and spectral transformation.

Jurečková et al. [19] (2024) developed a method that uses both classification and clustering techniques to handle malware and in the process was able to isolate zero-day malware. They use the EMBER dataset which contains around 400,000 labeled malware samples. At first, static analysis is used to extract features without executing the malware samples. These features and static attributes are then used to train a multilayer perceptron (MLP) classifier, a kind of feed-forward neural network. This is able to assign incoming samples to known malware families. For samples that cannot be confidently classified by the MLP are clustered into new groups using self-organizing maps (SOMs), a kind of neural network that is unsupervised. This allows the formation of new malware families (Jurečková et al., 2024, p. 8). Finally, the newly formed groups are analyzed and if conclusive are integrated into the overall malware detection framework and so that later, newly identified variants are easily classifiable.

Roopak et al. [21] propose an unsupervised intrusion detection system (IDS) to identify zero-day DDoS attacks in IoT networks without any previous ideas of the patterns of attack. The model proposed by them uses Gaussian Random Projection (GRP) for the reduction of feature space, which effectively lowers the feature space from 87 dimensions to 25 dimensions while still maintaining important structural relationships. The suggested ensemble method integrates three unsupervised classifiers—K-means, Gaussian Mixture Model (GMM), and Stochastic Gradient Descent-based One-Class SVM (SGD-OCSVM)—through the application of a hard voting technique to enhance classification robustness. The system was tested on the CIC-DDoS2019 dataset and yielded a very good accuracy of 94.5% and F1-score of 94.3%, which outperforms some state-of-the-art unsupervised methods. Roopak et al. note that the model can effectively handle high-dimensional, unlabeled data with efficiency, making the model amenable to real-time detection of new IoT-based attacks.

ML and DL approaches have recently become popular in dealing with Zero-day threats. Dahal et al. [17] (2024) in their paper also use ML and DL, specifically multilayer perceptrons (MLP), and Explainable AI (XAI) for extensive interpretability. They utilize the KDD99 dataset, which is a popular intrusion detection system dataset that contains about 4.9 million data samples and 41 various features classes into 23 classes. In order to counter class imbalance, the authors merged these 23 classes into 4 general categories: probe attacks, distributed denial of service (DDoS), unauthorized access, and normal. This data is used to train four different multilayer perceptrons, which include baseline—which was the standard model trained on original data, weighted-trained with class weights, truncated—a smaller model with fewer parameters, neurons and layers to improve training speed, and finally weighted truncated models—combining the truncated architecture with class weights to handle class imbalance. Furthermore, they use SHAP values from the XAI framework to better model prediction.

2.3 Summary of Key Findings

Although the proposed model by Brinkley et al. [16](2024), has strengths, the paper pointed out a number of resource optimization and scalability concerns such as the Neural Network model utilized a lot more CPU and memory resources during the test phase, and also added latency in detection. They suggested the use of lightweight Neural Networks or ensemble approaches to lower computational complexity. In addition, they proposed incorporating adaptive learning procedures that could enable the model to learn adaptively new ransomware activity in real time.

Satam et al. [13](2023) in their study highlighted some failings on detailed designs and practical implementations, particularly concerning the use of honeypots for data collection, which is a notable limitation. Despite this, the study offers valuable foundational knowledge on reinforcement learning (RL)-based approaches and illustrates how ML techniques are evolving in the cybersecurity domain. While the paper effectively focuses on detection techniques, it falls short in addressing real-time adaptation and dynamic response mechanisms for handling zero-day attacks.

Pitre et al. [7](2022) achieved practical usability by reducing false positives with a minimal increase in false negatives. Although outcome is encouraging, these results are limited to the CIC-IDS2018 dataset.

Zaki et al. [24](2024) highlighted IoT systems are at risk from more complex attacks like distributed denial-of-service (DDoS). However, because this paper only looks at IoT scenarios, it can't be applied to general-purpose networks. It also raises concerns regarding its generalizability to real-life settings in that it is based on certain datasets. Overall, this paper provides the use of ensemble methods for detecting zero-day attacks.

Kareem et al. [22](2024) achieved excellent accuracy but they have some limitations like computational complexity as transformers require significant computational power. They recommended to focus on optimizing the model for resource-constrained environments by using model quantization and knowledge distillation in future work.

Abbas-Escribano and Debar [8](2023) noted the limitation of this research was the relatively short exposure period and that the honeypot could not lure advanced attackers that much. Future work suggests extending the duration of the deployment, adjusting security measures to lure such attacks better, and improving the data collection mechanisms for more comprehensive analysis.

The challenges of high-dimensional data and the relevance of older datasets in the present threat landscapes were the limitations. Diloglu [3](2022) suggested putting

more focus on real-time data and integrating deep learning with other traditional methodologies (Bayesian optimization) to obtain maximum precision and efficiency of intrusion detection systems.

Balamurugan, M. [15](2024) highlighted the limitations of this article was the reliability on Artificial Intelligence brings in its own set of vulnerabilities such as different types of adversarial attacks are targeting different types of AI models. There is a huge challenge which remains as of now-reducing the false negatives further. While it has been tested in controlled environments, accessibility over many real-world connections is still a challenge. The next goals of the article are to enhance detection accuracy and flexibility for different network circumstances. Furthermore, a combination of IoT for wider threat coverage and blockchain technology for data security. And lastly, protection of privacy in collaborative cyberattacks using federated learning.

The proposed model by Igugu, A. [18](2024) became less successful when they had to identify unknown zero-day attacks. Models failed to adequately detect unexpected attacks and revealed issues with real-world applications. Based on the study, machine learning models, especially ensemble methods, are highly promising, yet they are still being developed to be capable of effectively thwarting zero-day attacks in changing cloud environments. Analysis of non-ML-based solutions or other cloud arrangements is given less thought.

Nkongolo et al. [6](2022) noted the limitation of this research was that individual classifiers (SVM and NB) are unstable and there are not enough data samples to improve the models. Future work suggests developing deeper networks to improve the accuracy of detection and to collect more datasets to extend the contribution of their approach.

Sarhan et al. [12](2023) records in their research paper that the model requires sufficient training data on known attacks for constructing the model which may not be available in all cases. The dynamic network environment may not be suitable for the model therefore may affect accuracy. This was not taken into account when building the model. Future works: Integration real time data can be tried and explored for enhancing the models capabilities over time. Other Machine Learning algorithms or even hybrid models can be designed based on the findings of this research. Furthermore, newer datasets can be explored for training.

Sayduzzaman et. al [23](2024) noted that in their research paper there are limitations such as the efficiency of the model largely depends on the data used to train the model, therefore if a smaller less diverse dataset is considered for training it may result in a lower performance model. Furthermore, the model has high computational time which may significantly increase further if assessing large quantities of data is required. Lastly, achieving true interoperability will remain a challenge due to current technological limitations.

Hairab et al [4] (2022) have discussed a few grave issues such as overfitting of the ML/DL model. These models perform well at training but do not identify unseen data. They have used the model's complexity or limited data with which a good model is created as the reason here. Another reason is the same distribution of the testing data and the training data. They have mentioned improving the time optimization techniques for CNN as future work in order to accelerate the training of CNN. They have another proposal to create other scenarios with different combinations of Botnet attacks with other diverse datasets.

In the article, Mbona et al [5] (2022) finds that among the shortcomings of Benford's law is that it is only applicable to positive real numbers, wherein all the digits from zero through nine can be observed. They stated for future work that they need to create a system that employs multiple feature selection methods and ML classifiers in a manner that high-performing approaches are integrated rather than adopting a single approach individually.

Kandhro et al. [11] demonstrates that deep learning architectures like CNNs and RNNs can effectively identify both conventional and zero-day threats in IoT environments, achieving high classification accuracy on standard datasets including NSL-KDD and UNSW-NB15. While the study proves the viability of real-time deep learning pipelines for threat detection, its practical application is constrained by two critical limitations: the absence of industrial IoT environment validation and lack of integration with honeypot-derived attack signatures. These gaps significantly reduce the system's ability to recognize real-world attack patterns, particularly novel threats emerging in operational technology contexts. Our work directly addresses these shortcomings by incorporating industrial honeypot data and combining deep learning with hybrid anomaly detection techniques, thereby extending the detection capability to previously unseen attack patterns while maintaining the real-time performance demonstrated in their study.

Faldi et al. [9](2023) demonstrates how well honeypot systems work at capturing well-known attack patterns like GoldenEye and Slowloris, and its pfSense/Snort IDS implementation is especially strong at identifying application-layer threats. The study sheds important light on real-time alarm generation in controlled conditions and the analysis of attacker behavior. But the primary drawback of their signature-based strategy is that it is unable to recognize new attack patterns, which is a crucial need for modern cybersecurity systems. Our research builds on their honeypot framework and improves this methodology by incorporating machine learning-based anomaly detection. This bridges the gap in zero-day threat identification by extending threat detection to previously unknown attack patterns through advanced behavioral analysis and pattern similarity evaluation, while also maintaining their real-time alert capabilities.

Ali et al [2](2022) emphasized on some known flaws in the field include stale datasets such as KDD99, high false-positive and false-negative counts, a focus on massive attacks, vague ways to measure success, dataset bias, and slow response times. New

criteria for real-time identifying, more precise metrics, mixed-method models, lower-alarm objectives, and implementation across IoT devices and associated cars are among the initiatives that remain to be accomplished. The evaluation emphasizes that any solution for sudden zero-day attacks needs to be strong; although Stacker and tDCGAN exhibit future potential they both have trouble with situations that are critical and low-volume incidents, therefore flexible A.I. techniques should continue.

Topçu et al [14] (2023) in his paper states that though it has incredible usefulness, the software has a variety of shortcomings, including over-reliance on keyword searching, technical constraints to real-time scaling, and the inability to continuously target individuals or groups of Twitter users. Further, human engagement may be necessitated for effective understanding of tweet importance. Given these constraints, the authors propose future work that will entail the addition of adaptive, self-learning models for dynamic threat trends, launching the application on additional social networks, and expanding real-time detection. In general, the research demonstrates a new strategy to active cybersecurity threat detection from machine learning and social media.

Despite this broad toolkit, Guo [10](2022) notes detection accuracy remains uneven, a problem tied to differing metrics and limitations in the datasets used. Deep-rooted issues-such as lack of standardization, fragmented collections, and thin feature engineering-still weaken model performance and make it harder to apply results across new settings. Incorporating domain knowledge in feature crafting, exploring novel machine learning strategies like reinforcement learning, and employing honeypots for collecting real data in future research. To facilitate fair and large-scale comparisons, the study strongly suggests creating a standard reference collection with heterogeneous, rich datasets and automatic evaluation protocols. Regarding realizing effective zero-day attack detection, the paper points out both the promise and immediate limitations of current machine learning approaches.

Jurečková et al. [19](2024) in their paper developed a model that enables online processing which brings dynamicity to the entire framework. Achieved classification accuracy in the first phase is 97.21% with balanced accuracy of 95.33%. Samples which could not be confidently classified, in the second phase, had to be clustered. The clustering achieved a clarity ranging from 47.61% (with four clusters) to 77.68% (with ten clusters).

Roopak et al.[21] in his paper states that the IDS model is good at detecting zero-day DDoS attacks, but the article also cites some drawbacks. The ensemble architecture, while improving accuracy, adds computational burden and complexity due to the coordination of numerous unsupervised models. Roopak et al. mention that maximum performance largely relies on proper hyperparameter tuning, which could interfere with scalability in real-world applications. To fill this gap, the authors propose investigating light-weight clustering techniques and integrating adaptive learning modes that can cause the system to learn and adapt from developing threats. They also

propose that in future, the framework can be extended to other types of cyberattacks, such as brute-force and XSS-based attacks, to make the framework even more effective.

In Dahal et al. [17] (2024), the truncated models were found to have fewer parameters than the general baseline model. The two abbreviated models performed better than baseline model and weighted model regarding accuracy, precision, recall and F1 scores. We used class weights in the models to account for the class imbalance issue. The reduced model fared best in accuracy (99.62%) and acceptable accuracy, recall. Although the accuracy of weighted truncated was somewhat lower (97.26%) but was more balanced in class detection to reduce the bias towards uncommon attack types.

Chapter 3

Proposed Methodology

3.1 Design Process or Methodology Overview

The ultimate objective of this research is to create and construct an integrated, adaptive, and smart intrusion detection and prevention system that not only detects traditional and known attacks, but is also capable of detecting and responding to zero-day attacks in real time. To achieve this, the design process is organized into six connected but separate stages: model training by dataset, IoT honeypot data consolidation, enhanced anomaly detection, adaptive defense augmented with reinforcement learning, integration with explainable AI, and ultimate model testing with real threats and simulated attacks.

1) Model Training by Dataset: The approach began with supervised Random Forest classifier training on the CICIDS2023 dataset. The classifier was selected since it is exceptionally good at categorizing known attack types, is noise-tolerant, and capable of working with imbalanced class distribution. It was the basis for detecting potential anomalies in live traffic.

2) Use of IoT Honeypot Dataset: In the meanwhile, IoT honeypot traffic was gathered, preprocessed, and normalized. It was real malware traffic that was intercepted in a virtual setup. This was necessary for presenting realistic attack patterns, independent of the training set. The objective was to compare new anomalies against this honeypot-based baseline in an effort to identify new threats.

3) Improved Anomaly Detection: The anomalies detected by the Random Forest model trained on the data were thoroughly analyzed. These anomalies were cross-matched with the honeypot data using Manhattan distance-based metrics. Any sample that didn't fall under recognized attack patterns was identified as an "un-

common anomaly.” These uncommon outliers are possible zero-day activities or threats that don’t fit into recognized attack patterns.

4) RL-driven Adaptive Defense: The second is to build an adaptive defense, layer through reinforcement learning. Deep Q-Networks (DQN) and Proximal Policy Optimization (PPO) algorithms will be utilized. These agents will acquire the ability to perform actions (e.g., IP blocking) based on feedback from the observed anomalies. In time, the system will automatically respond to routine malicious attacks. For instance, if an IP address is determined to be used recurrently for malicious activity, it will be blocked beforehand in the future.

5) Explainable AI Decisions: To promote transparency and trustworthiness, the system will incorporate an Explainable AI (XAI) module. This will generate readable and informative reports on the reasons why an event has been identified as anomalous, what features were responsible for the decision, and what impact it can have. For instance, where the system may provide context information such as: ”Identified brute force SSH login from IP X.X.X.X with common password patterns.” This will not only help analysts but also facilitate sharpening RL policies.

6) Model Testing: The last step will be testing the model for generalizability on known and unknown attacks. Known attacks will be drawn from benchmark data sets such as CICAPT-IIoT. Synthetically generated data will also be used through Generative Adversarial Networks (GANs) to mimic unknown, zero-day attacks. Detection performance of the system for such unseen forms of attacks will be the main test of generalization, resiliency, and practical usability in real life.

In short, the approach is not just crafted for detection but also for ongoing learning, flexibility, and interpretability. Though the present development has addressed the training, preprocessing, and improved anomaly filtering pipeline, reinforcement learning, explainability layer, and general model assessment are the next phases to be conducted in this research cycle.

3.2 Preliminary Design or Design (Model) Specification

We have evaluated 3 machine learning models and 1 deep learning for performance. These are -

- Random Forest
- XGBoost

- CatBoost
- MLP Classifier

Each model was trained on a preprocessed version of the CICIDS2023. There were 33 types of attack based on 7 categories. We choose 1 attack from each category and benign rows and from this we selected a certain number of rows for the training dataset to make it more balanced. Also we make a testing dataset by choosing a certain number of rows from each type of attack. Then we trained the model and tested it. Known attacks we marked and unknown attacks were labeled as zero days attack. These models were executed and contrasted for determining the best approach for the classification and prediction of zero-day attacks. The evaluation phase involved Predicting on the test dataset, Extracting anomalous predictions and Measuring classification accuracy, precision, recall, F1-score, and confusion matrices. A selection process of attributes phase was applied using Random Forest importance ranking to retain the top 20 most informative features. This reduced noise and improved model generalization. Hyperparameters for each model were selected through empirical tuning and grid search methods. For example, CatBoost used 300 iterations and depth=6, while MLP used two levels that are invisible with 100 and 50 neurons simultaneously. We used Python libraries for implementing all experiments such as Scikit-learn, XGBoost, CatBoost, and Matplotlib for visualization. Jupyter Notebooks and standalone scripts were used to track training and evaluation times.

Model	Accuracy	Macro F1	Precision	Recall
Random Forest	0.99	0.97	0.98	0.97
XGBoost	0.88	0.82	0.90	0.79
CatBoost	0.87	0.78	0.89	0.76
MLP Classifier	0.87	0.79	0.89	0.76

Table 3.1: Performance comparison of different models on standard classification metrics. Random Forest achieves the highest overall performance.

After a comprehensive comparison, Random Forest was selected for downstream anomaly detection and adaptive defense integration. The selection was based on some factors like -

- Performance : It Achieved the highest accuracy, macro F1-score, and near-perfect classification across all known classes
- Interpretability: Feature importance and decision logic are easily understandable, which is essential for security applications.
- Efficiency: Faster training time than other models and robustness against noise and overfitting

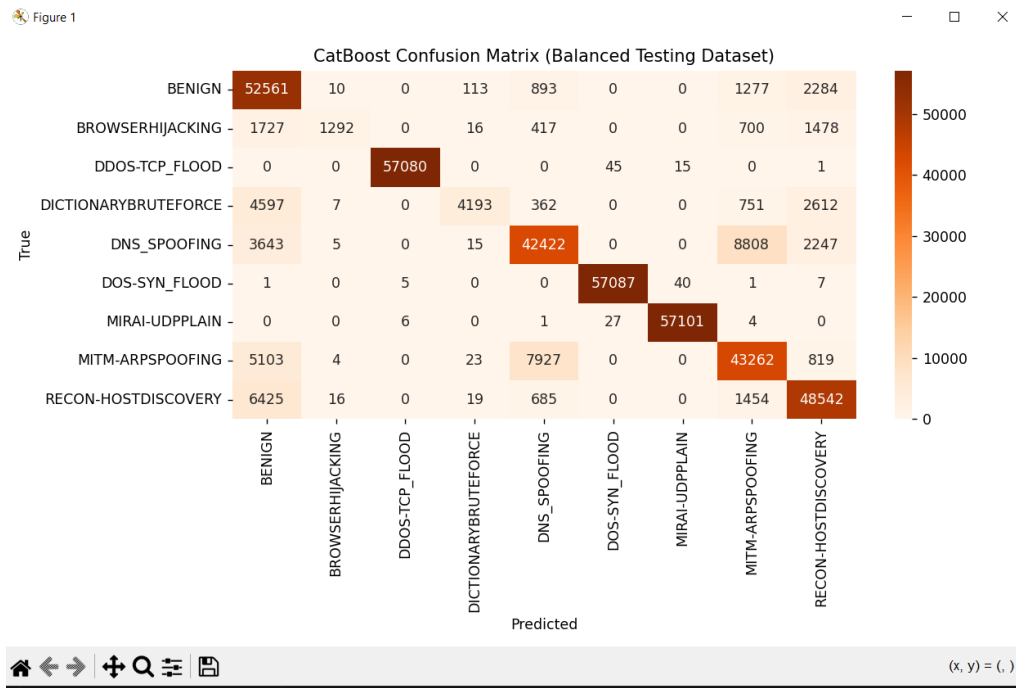


Figure 3.1: Confusion matrix for CatBoost classifier showing classification performance on known and zero-day attacks.

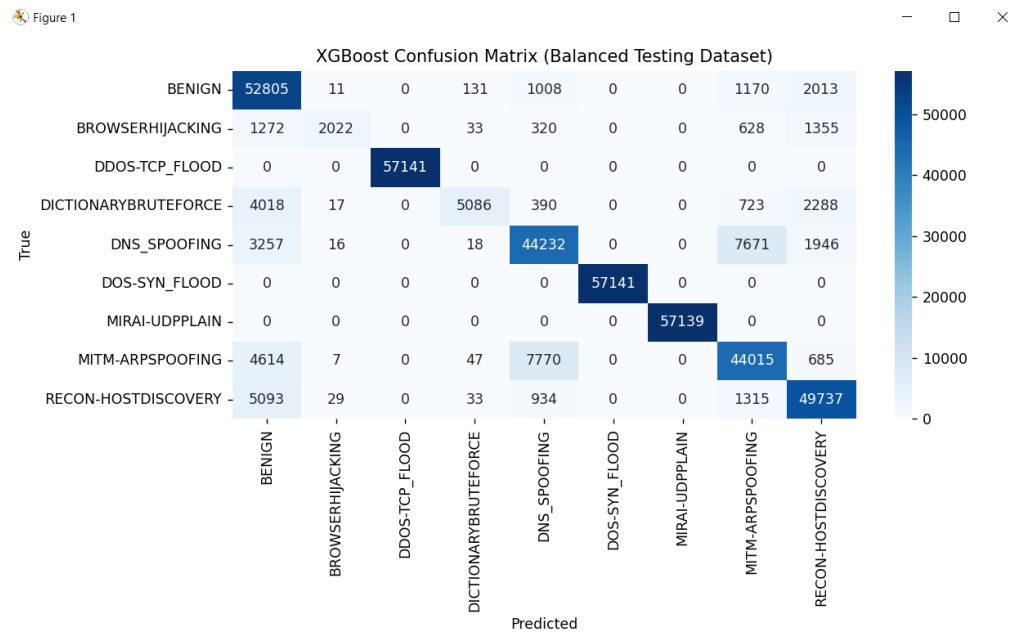


Figure 3.2: Confusion matrix for XGBoost classifier showing classification performance on known and zero-day attacks.

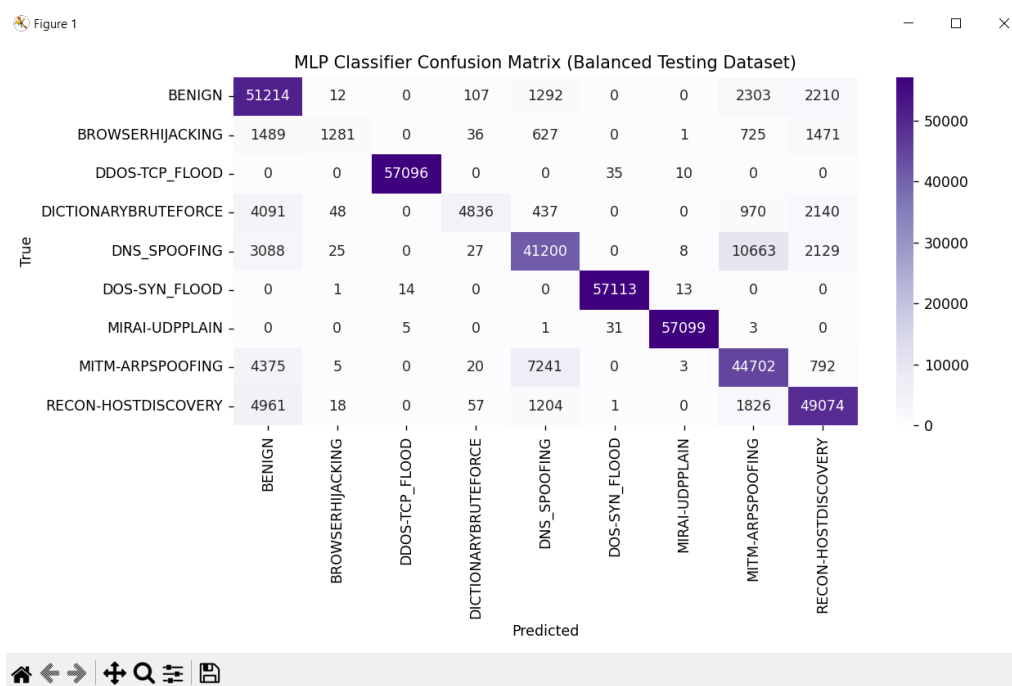


Figure 3.3: Confusion matrix for MLP Classifier showing classification performance on known and zero-day attacks.

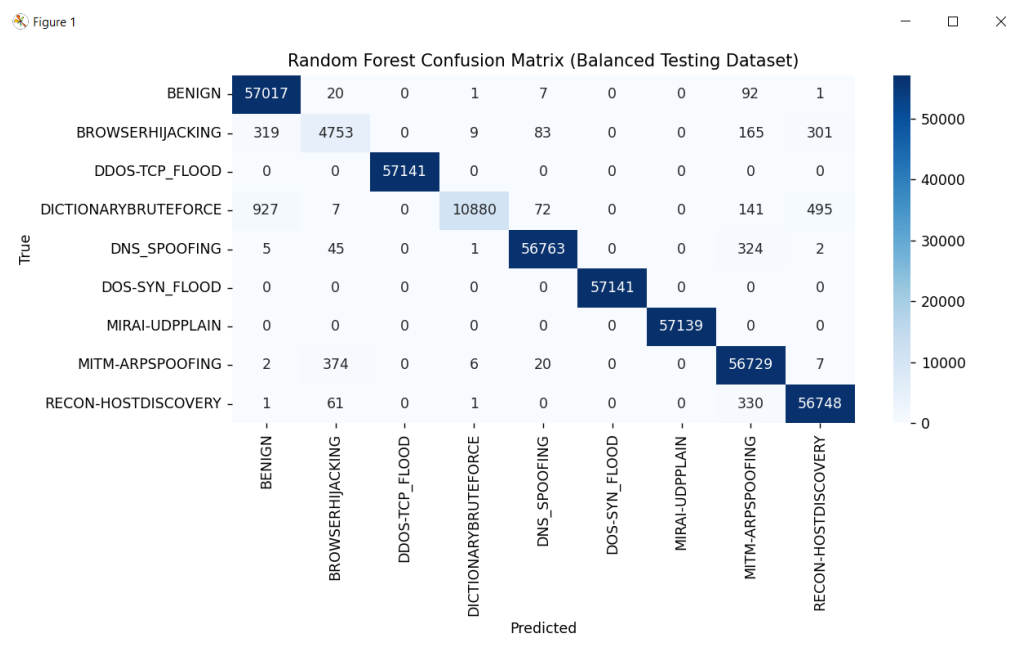


Figure 3.4: Confusion matrix for Random Forest classifier showing classification performance on known and zero-day attacks.

Also The Random Forest matrix demonstrated near-perfect classification for dominant classes like DDOS-TCP_FLOOD, DNS_SPOOFING, MIRAI-UDPPLAIN. On the other hand, a DL model was chosen for its capability to model non-linear relationships, especially in high-dimensional feature spaces. We selected these models because they combine traditional ensemble learning with modern deep learning techniques, giving us a well-rounded understanding of how different approaches perform against both familiar and emerging threats. While Random Forest offered excellent performance, it may not scale as efficiently with extremely large datasets. MLP-Classifer, although effective in learning non-linear patterns, was more sensitive to feature scaling and required longer training time.

3.3 Data Collection

We have used CICIoT2023 datasets as our primary dataset. This dataset was downloaded from the official Canadian Institute for Cybersecurity (CIC) repository, which is publicly accessible for academic and research purposes. It contains benign traffic and various types of attack including 11 types of DDOS attack, 1 type of brute force attack, 2 types of spoofing attack, 4 types of DOS attack, 5 types of Recon attack, 6 types of Web-based attack and 3 types of Mirai attack. Each attack is captured in a separate CSV file containing detailed network features such as flow statistics, packet information, TCP flags, and protocol behaviors. There were total of 63 individual CSV files were consolidated to form a unified dataset. We chose this dataset because it contains high variety of realistic attack types, relevant to modern IoT environments. It contains a comprehensive feature set, allowing for both supervised and unsupervised learning techniques. It has balanced inclusion of benign and malicious data. Another reason behind choosing this dataset was public availability and wide adoption in recent academic research which makes it a reproducible and credible benchmark.

Apart from the CICIoT2023 dataset, we also used another honeypot-based dataset namely—the IoT-DH Dataset, which was released on Mendeley Data in January 2024 by the Universitas Muhammadiyah Malang and Universitas Gadjah Mada authors. The dataset is a customized dataset that is specifically tailored to improve the knowledge of Distributed Denial of Service (DDoS) attacks on Internet of Things (IoT) networks. Seized in an isolated honeypot environment, it consists of genuine network traffic, packet captures (PCAP), and system logs that emulate real attacker patterns. The data attempts to expose adversaries’ plans, strategies, and procedures utilized to overwhelm IoT systems. For easy analysis, it was first captured in PCAP format and then changed to CSV with 10 primary features like dt, dur, dur_nsec, tot_dur, pktrate, protocol, port_no, tx_kbps, rx_kbps, tot_kbps, and a labels column. This cleanly parsed and labeled dataset was useful to us for our analysis, especially for cross-validating and comparing the anomalies identified through our supervised models. Its actual malicious traffic, responsiveness to the current DDoS threats, and recent soundness in research rendered it a valid measure for our honeypot module

in zero-day detection studies.

3.3.1 Data Cleaning

We used several techniques for data cleaning. First of all, machine learning algorithms can't process infinite values which appear during feature extraction or computation. We replaced these infinite values with NaN so that they can be handled uniformly. Machine learning algorithms also can't handle missing values (NaN) if they are not specifically supported. We removed those rows containing missing values. We removed this after replacing Inf. This ensures that the training dataset contains valid and complete only. This also reduces our dataset size. We also handled unknown labels. We separated zero days attack types during testing which is unknown to the model. If any labels are not seen during training, it is marked with -1 and saved as anomalies during testing. This is not a traditional data cleaning step but it is important to our quality control. We also used z-scores to find and select values that are significant.. We also Identified and removed duplicate rows. We used z-scores to find and select values that are significant.. As we make the training dataset as balanced as possible , we didn't need extra processes to make it more balanced. If we use larger dataset in future steps of our work, we will use some other data cleaning method like Winsorization which ap extreme values to a specific percentile instead of removing outliers, Feature Variance Thresholding which will drop features with near-zero variance, SMOTE (Synthetic Minority Oversampling Technique) if dataset is imbalance, Anomaly Score-Based Cleaning which will assign anomaly scores (e.g., using Isolation Forest) to the training data itself and remove very high-score points (outliers)

3.3.2 Data Transformation

We used label encoding to convert string-based class labels like "BENIGN", "DOS-SYN_FLOOD" into numeric format for our supervised models. It is effective for models like XGBoost and Random forest which expects numeric value. Every category receives a distinct integer by label encoding. Labels like "BENIGN": 0, "DOS-SYN_FLOOD": 1, "MITM-ARPSPOOFING": 2, ... 1. We also used standard scaler for feature scaling. After subtracting the average and divides by standard deviation. Attributes get small variances and zero average as a result. It is so important as it ensures features with large magnitude like Tot_sum, Tot_size don't dominate other features. It improves the overall performance of models like MLPClassifier, SVM that are sensitive to feature scale. It ensures that all input attributes provide the same impact to the procedure of understanding In some experiments, we did feature selection using model-based methods, such as feature importance scores from tree-based classifiers. This reduced dimensionality and improved training efficiency by choosing the top N most applicable attributes. The selection of attributes also

helped to minimize overfitting and decreased noise from irrelevant or redundant features.

3.3.3 Data Integration

Data integration in this study comprised the conscious combination of two different but related datasets: one produced by supervised anomaly detection algorithms (Random Forest specifically), and another obtained from an IoT honeypot setup. The datasets differed naturally in structure, source, and purpose. The former (Random Forest anomalies) was based on standard machine learning classification on a labeled data set (e.g., CICIDS2023), whereas the latter was real, actual, unsupervised threat data collected by a honeypot system that is meant to attract and collect malicious traffic. The process of integration started with the creation of a common feature space between these two data sources. After domain knowledge and schema matching, the common features were determined and normalized in both data sets. More specifically, Rate, Tot size, IAT, Protocol Type, and Label columns were selected as common columns. These features depict important network flow properties like packet rate, byte total count, inter-arrival time, and protocol types. Basically, the features that are widely utilized in intrusion detection and anomaly type classification tasks. The Protocol Type field needed further normalization for inter-dataset compatibility. Protocol values were coded with a numeric convention (ICMP - 1, TCP - 6, UDP - 17) in the honeypot dataset to match the coding utilized or anticipated in the model-detected dataset. Integration was performed cautiously by making sure that both datasets had passed through the same preprocessing steps like removal of missing values, enforcement of numeric types, and Min-Max scaling so that the resulting final combined feature space can be easily comparable in subsequent steps, especially while distance-based anomaly matching. This step was important in permitting cross-source anomaly comparison, so that the output of the trained model could be intelligently compared to actual-world honeypot activity, enabling the ultimate detection of rare or zero-day anomalies.

3.3.4 Data Reduction

Data reduction in this case was done carefully to achieve feature space consistency and computational efficiency, particularly with respect to the volume of data processed in batches. Although no approaches to decrease complexity such as PCA (Principal Component Analysis) were used as such, feature selection by hand was instrumental in reducing the complexity of the analysis without compromising any relevant signal information. Feature selection was based on domain knowledge and pragmatics. Just a small subset of high-information features (Rate, Tot size, IAT, and Protocol Type) from both datasets were kept. This not only amounted to compatibility between datasets but also reduced the dimensional complexity of sub-

sequent distance calculations. These features were empirically selected on the basis of their significance in describing the traffic volume, time patterns, and protocol utilization, which are all valuable information in anomaly detection contexts. Moreover, reduction in the dataset size was obtained by removing the non-anomalous data. Only the rows labeled 1 (attack) from the honeypot dataset were kept, since these records are most comparable to the anomalies detected by the trained model. Similarly, in the final distance-based filtering, only some of the model anomalies that had exhibited high divergence from the honeypot behavior (based on Manhattan distance thresholds) were kept for analysis. Distance-based filtering was a second-level reduction mechanism that focused on selecting out unusual anomalies, which are of greatest interest in zero-day detection scenarios. Hence, the coming together of selective column retention, label-based filtering, and statistical thresholding formed a unified data reduction pipeline that was not only computationally tractable but also analytically prioritized.

3.3.5 Summary of Preprocessed Data

Following extensive preprocessing and integration, two preprocessed and fully normalized cleaned datasets were obtained: `rf_anomalies_preprocessed.csv` and `honeypot_cleaned_normalized.csv`. The two datasets were standardized to a common format and structure, having the normalized columns: Rate, Tot size, IAT, Protocol Type, and Label. This preprocessing phase included dealing with missing and inconsistent values, Min-Max normalization of all numeric features, mapping protocol types to their respective numeric values (i.e., ICMP to 1, TCP to 6, UDP to 17), and encoding all the labels to binary, where benign samples are 0 and malicious samples are 1.

The `rf_anomalies_preprocessed.csv` file contains samples of network traffic that were identified as anomalies by a supervised Random Forest classifier, trained on the CIDS2023 dataset. The samples are of various known types of attacks, and their features have been normalized for consistency with the honeypot dataset. The `honeypot_cleaned_normalized.csv` file, however, contains actual malicious traffic captured from an IoT honeypot system. This data was batch processed to enable us to handle large quantities well and to retain for examination only records labeled as attacks.

After preprocessing, both sets were mapped onto shared feature columns and readied for Manhattan distance comparison. This allowed for the substantive examination of the degree to which model-identified anomalies approximated or diverged from real malicious activity recorded in honeypot logs. The final outcome of this comparison, comprised of anomalies that are themselves relatively unique from known attack patterns—thus possibly new or zero-day attacks—was recorded as `final_uncommon_anomalies.csv`. This data forms a significant part of the overall methodology by providing an added layer of anomaly filtering designed to draw

attention to traffic that does not match known threat profiles.

3.4 Implementation of Selected Design

The chosen design was made operational in a multi-phase pipeline that integrated supervised learning, honeypot integration, and statistical anomaly filtering, as illustrated in the workflow diagram of the project.

1) Model Training and Anomaly Extraction: A Random Forest classifier was trained using a labeled dataset (CICIDS2023) to label attack behaviors from some known attack categories. The outcome, `Anomalies_Random_Forest.csv`, was predictions labeled as non-benign and represented the model's list of anomalies.

2) Honeypot Dataset Preprocessing: A real IoT honeypot dataset was normalized, cleaned, and filtered to contain only malicious records only. Chunk-wise loading and processing strategy (using pandas and MinMaxScaler) allowed the easy processing of the massive dataset, giving rise to `honeypot_cleaned_normalized.csv`.

3) Anomaly Comparison by Distance Metrics: Every set was projected into the same feature space. Manhattan (cityblock) distance between every model-identified anomaly and honeypot entry was computed using `scipy.spatial.distance.cdist`, processed in decent-sized batches to keep within memory. For each anomaly, the closest distance to any honeypot attack was computed.

4) Thresholding for Uncommon Anomalies A distance histogram was plotted to watch for the distribution. Experimentally, the 80th percentile cutoff was chosen to remove the majority of the most dissimilar anomalies from honeypot attacks. These rows would most likely be anomalous or zero-day attacks since they are not associated with the behavior of known malicious patterns.

5) Saving Final Results: The infrequent filtered anomalies were saved to as "final uncommon anomalies.csv" for analysis, reinforcement learning incorporation, or adaptation in defense policy.

This linked architecture guarantees that the model can not only identify suspected risks but also label anomaly patterns that could represent new or developing attack patterns, thereby extending intrusion detection capacity in changing network environments.

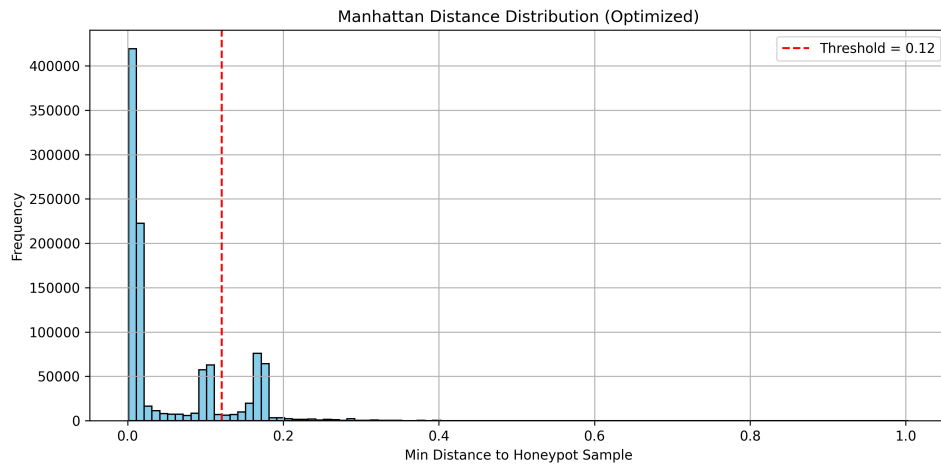


Figure 3.5: Manhattan Distance Distribution of test samples against honeypot dataset. A threshold of 0.12 (indicated by the red dashed line) was determined to isolate samples with uncommon behavior. Values above the threshold may indicate potential zero-day or novel attacks.

Chapter 4

Conclusion

The increasing threat posed by zero-day vulnerabilities emphasizes the need for a transformative approach towards cybersecurity that moves beyond traditional detection methods. Conventional signature-based solutions, IDPS systems are proving inadequate in detecting complex attacks and novel evasion techniques. We propose a hybrid framework leveraging artificial intelligence (AI), honeypots, Deep Q-Networks (DQN), Reinforcement Learning (RL), Convolutional neural networks (CNNs) and Explainable AI (XAI) to better detection capabilities and provide insights regarding behavior of the said attacks. Honeypot integration offers the primary intervention through the collection of attacker data. Convolutional neural networks (CNNs) can then analyze and filter the large datasets. Using Explainable Artificial Intelligence (XAI) transparency and clarity in the detection systems and decision making process is achieved. With this paper we hope to introduce the theoretical advancements in ML and AI in the application of a threat detection in the domain of cybersecurity hoping to design a resilient defense system against zero-day attacks, with significantly improved detection accuracy. Our proposed framework should enhance the operational efficiency as well as reduce false positives. Through our efforts we hope to design a reliable security system using which organizations will be protected and more capable to anticipate, predict, and reply to zero-day attacks.

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